

Q1: Smart City Surveillance System

Identify

- System: Surveillance (Video Processing)
- Components: CPU, Memory, I/O Devices
- Issue: Lag & delayed threat detection

Analyze

- Continuous data from cameras heavy I/O load
- CPU waits for memory bottleneck
- Single bus only one transfer at a time

Compare (Bus Structure):

Feature	Single Bus	Multi Bus
Speed	Slow	Fast
Data Transfer	One at a time	Parallel
Cost	Low	High
Efficiency	Low	High

Evaluate

- System performance poor due to:
 - o Bus congestion
 - o CPU idle time
 - o Memory delay

Recommend / Design

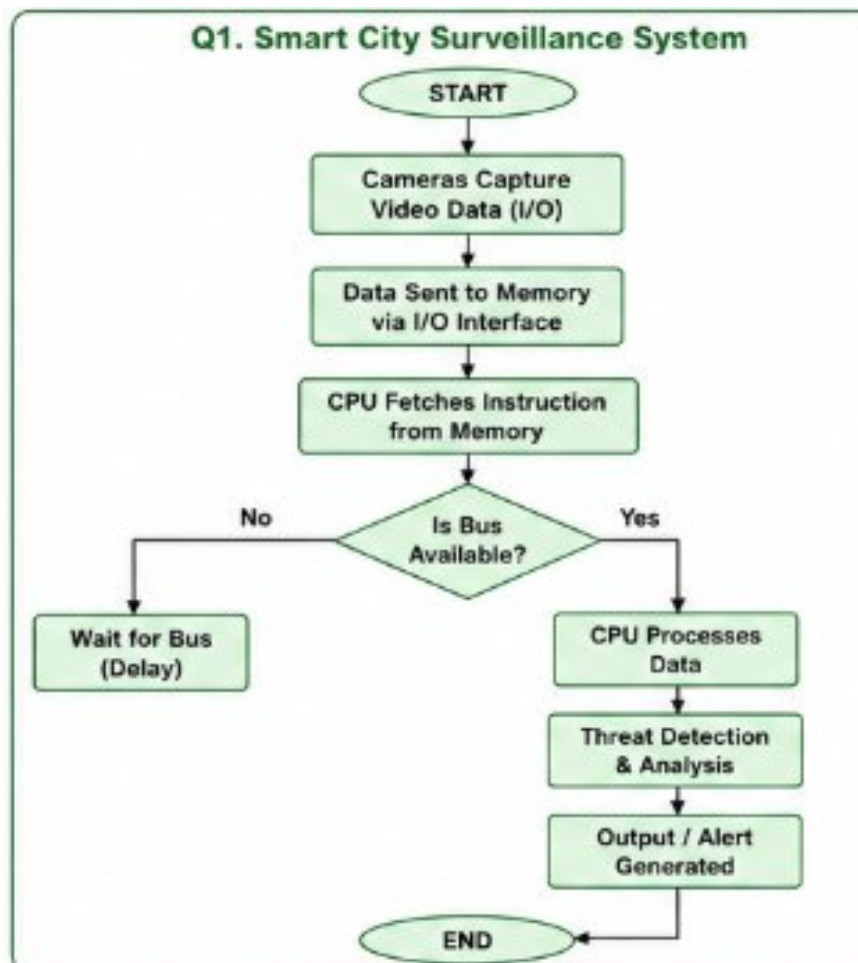
- Use Multi-bus architecture
- Add Cache Memory

- Use DMA (Direct Memory Access)
- Apply Pipelining

Justify

- Multi-bus reduces traffic
- Cache reduces memory access
- DMA bypasses CPU load

Flowchart



Q2: High CPI in Stock Trading System

Identify

- Issue: High CPI
- Cause: Memory access + Branch instructions

Analyze

- Memory delay increases cycles
- Branch pipeline flush
- CPU stalls frequently

Compare

Factor	Without Optimization	With Optimization
CPI	High	Low
Speed	Slow	Fast
Efficiency	Low	High

Calculate

Execution Time = CPI × Instruction Count × Clock Cycle

Example:

CPI = 4, IC = 1M, Clock = 1ns

Time = $4 \times 10^6 \times 1\text{ns} = 4\text{ ms}$

Evaluate

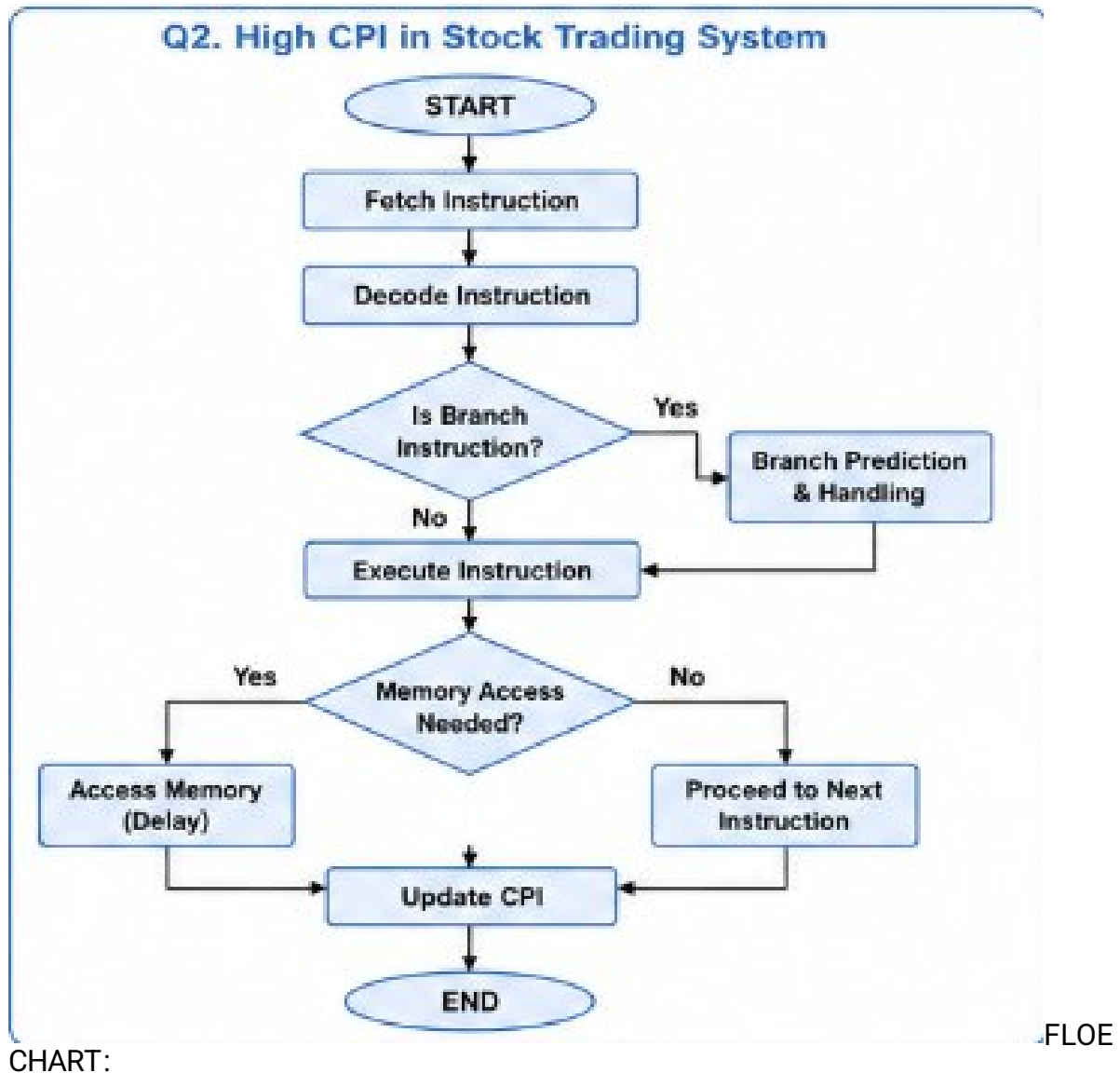
- High CPI reduces throughput
- Not suitable for real-time systems

Recommend

- Use Cache
- Apply Branch Prediction
- Use Pipeline Architecture

Justify

- Cache reduces memory latency
- Prediction avoids stalls



Q3: 8085 Addressing Mode Error

Identify

- Issue: Wrong temperature readings
- Cause: Incorrect addressing mode

Analyze

- Wrong address wrong data fetch
- Indirect errors sensor mismatch

COMPARE:

Mode	Example	Use Case
Immediate	MVI A,10H	Fixed data
Direct	LDA 2050H	Memory access
Register	MOV A,B	Fast operation
Indirect	MOV A,M	Pointer access

Evaluate

- Incorrect addressing system failure
- Industrial system requires accuracy

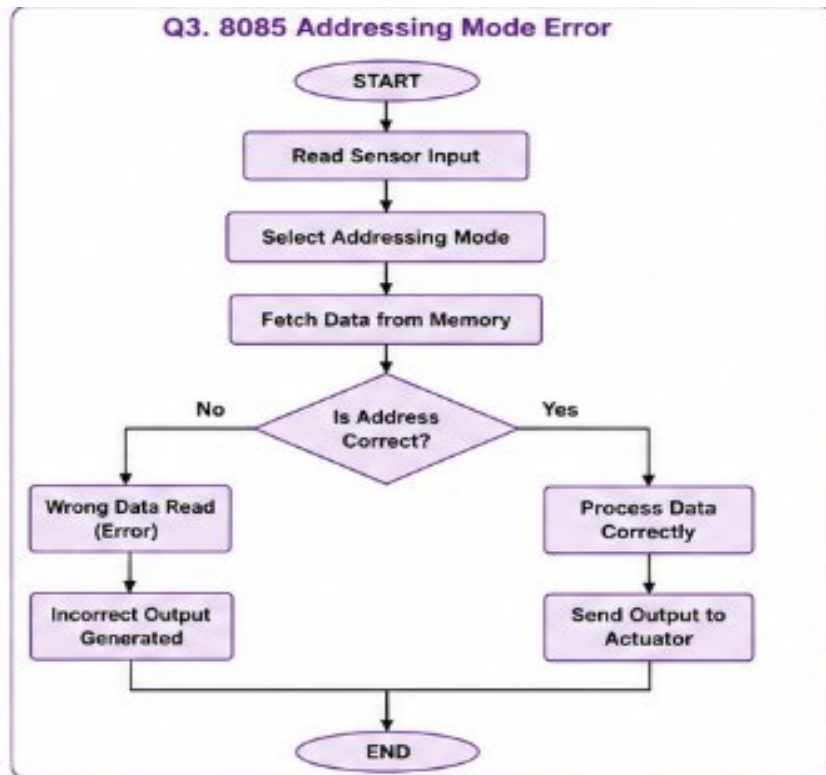
Recommend

- Use Direct addressing for sensors
- Validate memory addresses
- Debug using test values

Justify

- Direct mode ensures correct memory access

COMPARE:



Q4: 8086 Segmentation Problem

Identify

- Issue: Incorrect data access & stack overflow

Analyze

- Wrong segment registers
- Stack mismanagement

COMPARE:

Segment

Function

Code

Instructions

Data

Variables

Stack

Function calls

Evaluate

- Segmentation improves memory use
- Misuse serious errors

Recommend

- Proper segment initialization

Q5: Number System Conversion Error

Identify

- Issue: Wrong hex interpretation

Analyze

- Hex Binary error wrong instruction
- Data misinterpretation

COMPARE:

System	Base	Example
Binary	2	1010
Hex	16	A

Calculate

Example:

Hex = A3

Binary = 1010 0011
Evaluate

- Conversion errors wrong execution

Recommend

- Use automated conversion tools
- Validate data

Justify

- Correct conversion ensures accurate processing

FLOWCHART:

Q5. Number System Conversion Error

